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DIGITAL LOGIC

Assignment of Chapter Four

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A. FILL-IN-BLANK

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1. Each flip-flop consists of 2 (how many) steady states; each flip-flop can save 1 (how many) bits of binary number. $Q^{(n+1)} = T\bar{Q} + \bar{T}Q$
2. the next state equation of the T flip-flop is $Q^{(n+1)} = T\bar{Q} + \bar{T}Q$.
3. A basic flip-flop device made of NAND gates cannot have the RS=00 as its input.
4. T flip-flop, D flip-flop and JK flip-flops these three clock-controlled flip-flops have no constraints on input.
5. With the clock pulse on, in order to enable a J-K flip-flop to have $Q^{n+1} = \bar{Q}$, thus, the input signal must be J= 1, K= 1.
6. Once the clock pulse is on, a flip-flop's input can affect the next state of the flip-flop. This statement is true. (true/false)
7. A combinational logic circuit must not have any memory devices. This statement is true. (true/false)
8. 4 flip-flop devices can save up to 2 bits binary numbers. This statement is false. (true/false)
9. A basic R-S flip-flop made of NOR gates cannot have RS=00, such inputs. This statement is false. (true/false)
10. A clock based R-S flip-flop cannot have the situation that its both outputs are equivalent. This statement is false. (true/false)

B. SHORT ANSWER

3

- 1 What is the difference in the logic functions between a basic R-S flip-flop made of NAND gates and a basic R-S flip-flop made of NOR gates?

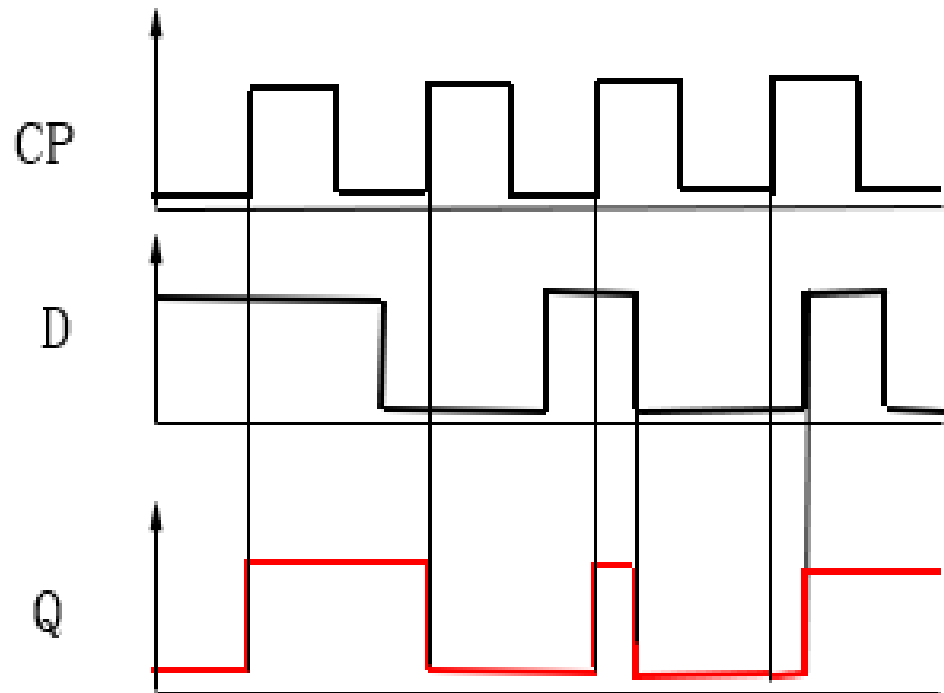
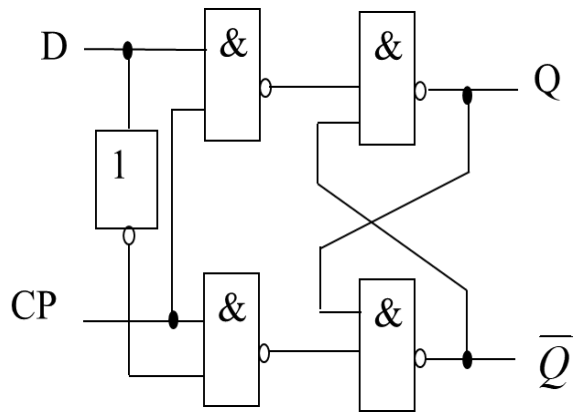
Basic R-S flip-flop function table

RS	Q^{n+1}	functional description
0 0	d	indefinite
0 1	0	set 0
1 0	1	set 1
1 1	Q	retain

Basic R-S flip-flop function table

RS	Q^{n+1}	functional description
0 0	Q	constant
0 1	1	set 1
1 0	0	set 0
1 1	d	indefinite

- 2 Fig 1 illustrate a D flip-flop diagram, and its input waveform is shown in Fig 2, please draw the output waveform of Q. (Assume the present state of Q is 0)





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Tutorial 5.1

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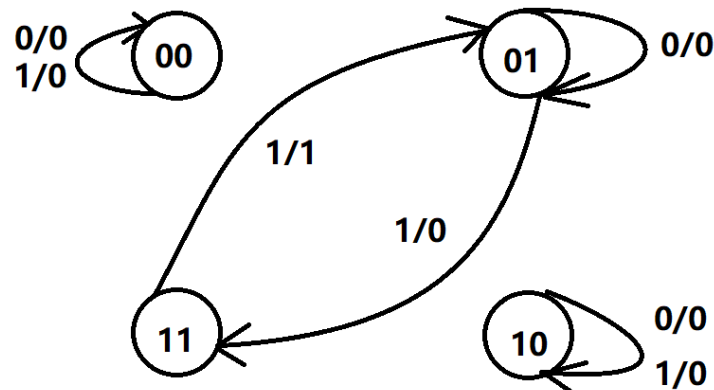
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- 1. Based on whether a sequential logic circuit contains a unified clock pulse generator, the sequential logic circuit can be categorized into synchronous- and asynchronous such two types.
- 2. A synchronous sequential logic circuit can be represented by Output function , excitation function , and next state function such three sets of logic function expressions.
- 3. The output function of the Mealy type sequential logic circuit consists of states+input , the output function of the Moore type sequential logic circuit consists of states .
- 4. “When the clock pulse is at a high-level voltage, the input signals can affect the states of the flip-flops”. This statement is true (True/False).
- 5. “The Mealy type and Moore type logic circuits are equivalent to each other.” This statement is true (True/False).
- 6. “The basic RS flip-flop can be directly used as the synchronous sequential logic circuit’s storage circuit. “ This statement is false (True/False).
- 7. “If the CP inputs of all the flip-flops in a sequential logic circuit are connected together, the circuit is the synchronous sequential logic circuit”. This statement is false (True/False).

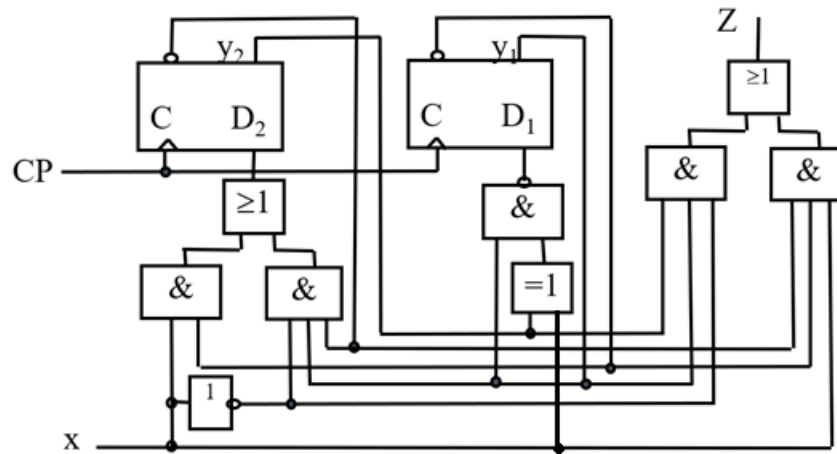
- Mealy Type, since one present state refers to two types of output:

Input	Present state	Next state	Output	Input	Present state	Next state	Output
X	$Q_2^s Q_1^s$	$Q_2^{s+1} Q_1^{s+1}$	Z	X	$Q_2^s Q_1^s$	$Q_2^{s+1} Q_1^{s+1}$	Z
0	0 0	0 0	0	1	0 0	0 0	0
0	0 1	0 1	0	1	0 1	1 1	0
0	1 0	1 0	0	1	1 0	1 0	0
0	1 1	1 1	0	1	1 1	0 1	1



Analyze the sequential logic circuit as shown in the following figure to illustrate the function of the circuit

4



A:

(1) Because the output of the circuit is related to the state and input of the circuit, this circuit is a Mealy type circuit; its output function and excitation function are::

$$Z = \overline{x}\overline{y_1}y_2 + \overline{x}y_1y_2$$

$$D_2 = \overline{x}\overline{y_1} + \overline{x}y_2y_1$$

$$D_1 = \overline{y_1}(x \oplus y_2)$$

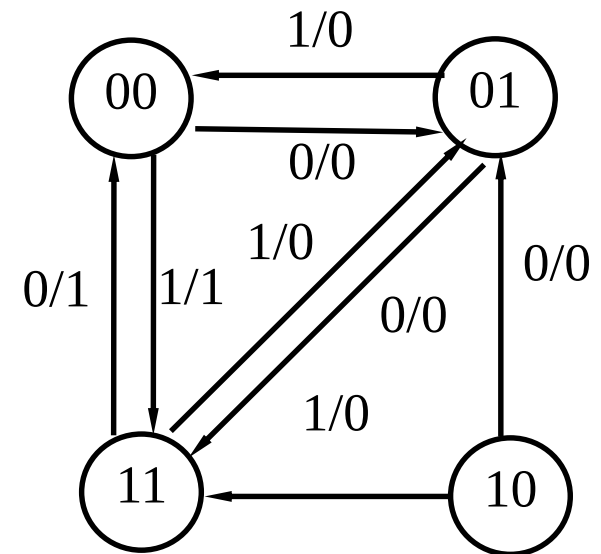
□ 2, Next state truth table

$$Z = x\overline{y_1}\overline{y_2} + \overline{x}y_1y_2 \quad D_2 = x\overline{y_1} + \overline{x}\overline{y_2}y_1 \quad D_1 = \overline{y_1}(x \oplus y_2)$$

Input X	Present y ₂ y ₁		Excitation D ₂ D ₁		Next state y ₂ ⁽ⁿ⁺¹⁾ y ₁ ⁽ⁿ⁺¹⁾		Output Z
0	0	0	0	1	0	1	0
0	0	1	1	1	1	1	0
0	1	0	0	1	0	1	0
0	1	1	0	0	0	0	1
1	0	0	1	1	1	1	1
1	0	1	0	0	0	0	0
1	1	0	1	1	1	1	0
1	1	1	0	1	0	1	0

3. State table and diagram

y_2y_1	$y_2^{(n+1)} y_1^{(n+1)} / Z$	
	X=0	x=1
00	01/0	11/1
01	11/0	00/0
10	01/0	11/0
11	00/1	01/0



- 4., This circuit is a ternary reversible counter. When $x=0$, an increment of 1 is achieved; when $x=1$, a count of minus 1 is achieved.

Analyze the synchronous sequential logic circuit as shown in the figure to illustrate the function of the circuit.

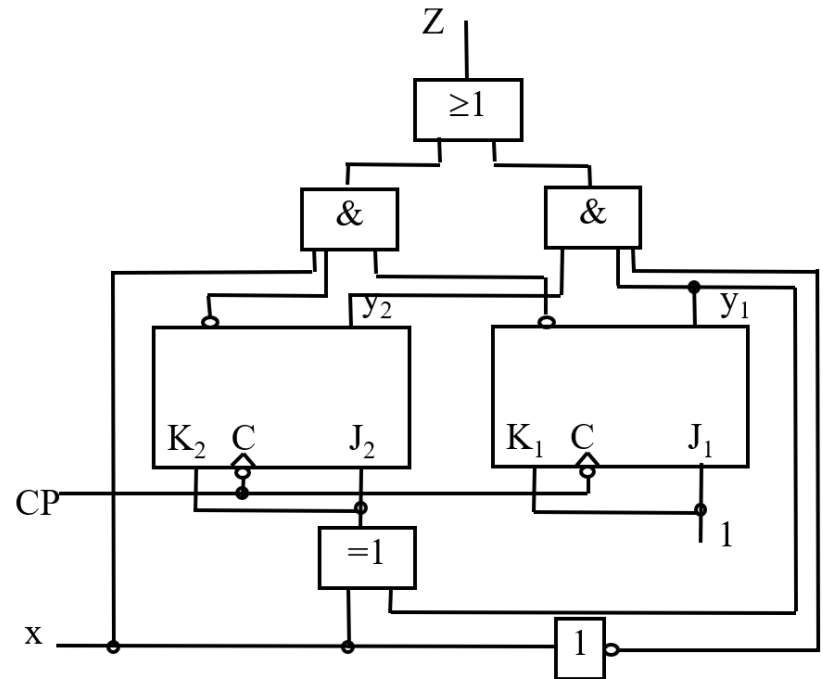
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- Because the output of the circuit is related to the state and input of the circuit, this circuit is a Mealy type circuit; its output function and excitation function are:

$$Z = x\overline{y_1}\overline{y_2} + \overline{x}y_1y_2$$

$$J_2 = k_2 = x \oplus y_1$$

$$J_1 = K_1 = 1$$



□ 2,Next State truth table

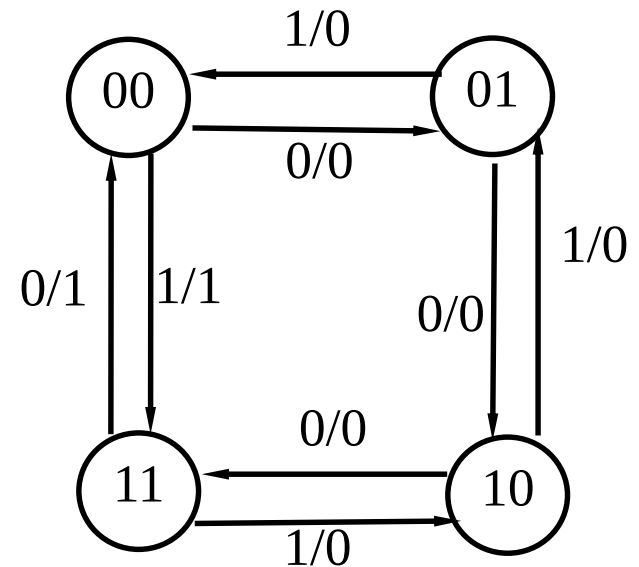
$$Z = x\overline{y_1}\overline{y_2} + \overline{x}y_1y_2 \quad J_2 = k_2 = x \oplus y_1 \quad J_1 = K_1 = 1$$

input X	Present		Excitation				Next state		Output Z
	y_2	y_1	J_2	K_2	J_1	K_1	$y_2^{(n+1)}$	$y_1^{(n+1)}$	
0	0	0	0	0	1	1	0	1	0
0	0	1	1	1	1	1	1	0	0
0	1	0	0	0	1	1	1	1	0
0	1	1	1	1	1	1	0	0	1
1	0	0	1	1	1	1	1	1	1
1	0	1	0	0	1	1	0	0	0
1	1	0	1	1	1	1	0	1	0
1	1	1	0	0	1	1	1	0	0

J	K	$Q^{(n+1)}$
0	0	Q
0	1	0
1	0	1
1	1	$\overline{Q}$

3 State table and diagram

$y_2 y_1$	$y_2^{(n+1)} y_1^{(n+1)} / Z$	
	X=0	x=1
00	01/0	11/1
01	10/0	00/0
10	11/0	01/0
11	00/1	10/0



(4) Functional analysis

- This circuit is a Mode 4 reversible counter. When $x=0$, the two-digit binary number is incremented by one, and the output is a carry signal; when $x=1$, the two-digit binary number is counted down by one, and the output is a borrow signal.



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Tutorial 5.2

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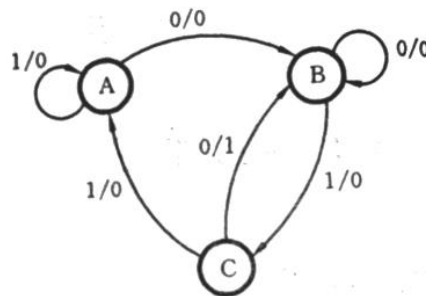
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1. The reduction of the basic state table applies the concept of equalivent.
2. Let the number of states contained in the modified state table be n , and the number of flip-flops in the corresponding circuit is m , then the relationship between m and n should satisfy $2^{m-1} < n \leq 2^m$.
3. Assume there is a Mealy type “0011” sequence detector, its modified state table consists of 4 (how many) states, and its circuit contains 2 (how many) flip-flops.
4. The state table of a synchronous sequential logic circuit is shown in the following table. If the initial state of the circuit is A and the input sequence of the circuit is $x=010101$, the corresponding output sequence of the circuit is 001100.

Present State	Next state/Output	
	x = 0	x = 1
A	B/0	C/1
B	C/1	B/0
C	A/0	A/1

3. Assume there is a Mealy type “0011” sequence detector, its modified state table consists of 4 (how many) states, and its circuit contains 2 (how many) flip-flops.

5. The state diagram of a synchronous sequential logic circuit is shown in the following figure. If the initial state of the circuit is A, and under the input sequence 110101010, the corresponding output sequence of the circuit is 0000 10101.



6. “In terms of the following equivalent pairs (A, B), (C, D) and (B, D), a maximum equivalent class {A, B, D} can be obtained.” This statement is (false)
7. To design a sequential logic circuit, the determination of the output function expression from the modified binary state table is independent of the type of the selected flip-flop. This statement is (true).

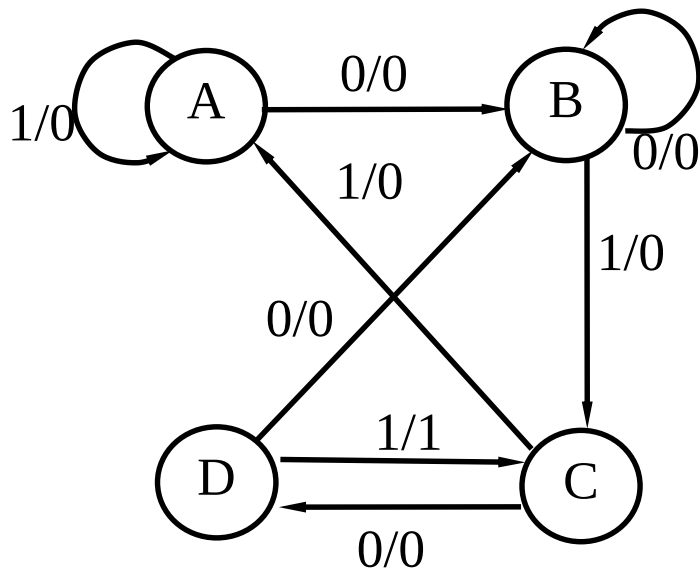
Section B

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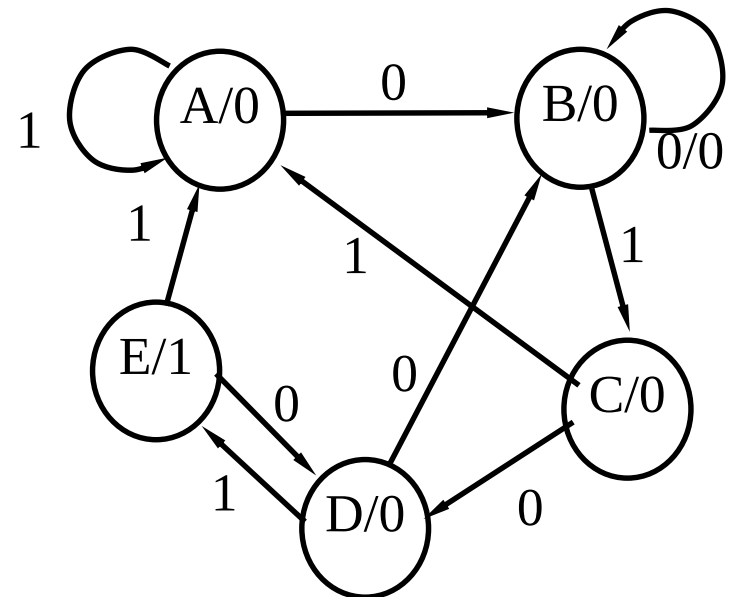
- Draw the Mealy and Moore type's basic state diagram of the "0101" sequence detector. 1. A typical input and output sequence is as follows:
- Input x: 1 1 0 1 0 1 0 1 0 0 1 1
- Output Z: 0 0 0 0 0 1 0 1 0 0 0 0

A:

(1) Mealy



(2) Moore



2 Reduce the basic state table

6

A:

Present state	Next state/Output	
	X=0	X=1
A	B/0	C/0
B	A/0	F/0
C	F/0	G/0
D	A/0	C/0
E	A/0	A/1
F	C/0	E/0
G	A/0	B/1

B	CF					
C	BE CG	AF FG				
D	AB	FC	AE CG			
E	×	×	×	×		
F	BC CE	AG FE	GE	AC CE	×	
G	×	×	×	×	AB	×
	A	B	C	D	E	F



(2) Maximum equivalent class
: {A,B,D}, {C,F}, {G,E}

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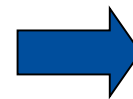
B	CF					
C	BF CG	AF FG				
D	AB	FC	AF CG			
E	×	×	×	×		
F	BC CE	AC FE	GE	AC CE	×	
G	×	×	×	×	AB	×
	A	B	C	D	E	G

Pres ent	Next/Output	
	X=0	X=1
A	B/0	C/0
B	A/0	F/0
C	F/0	G/0
D	A/0	C/0
E	A/0	A/1
F	C/0	E/0
G	A/0	B/1



pres ent	Next/Output	
	X=0	X=1
w	w/0	y/0
y	y/0	z/0
z	w/0	w/1

(3) Obtain the modified state table:
use w, y, z to represent {A,B,D},
{C,F}, {G,E}, then we have



33. Use the clock-controlled D, T, and J-K flip-flops as the storage devices of a synchronous sequential circuit, respectively, to implement the functions of the binary state table shown in the following table. Try to write the excitation function and the output function expressions; and try to answer “which type of flip-flop can achieve the simplest circuit?”

Present	Next/Output	
	X=0	X=1
00	01/0	10/0
01	11/0	10/0
11	10/1	01/0
10	00/1	11/1

A: (1) YZ matrices

x	y_2y_1	$y_2^{(n+1)}y_1^{(n+1)}$	D_2D_1	T_2T_1	$J_2K_2J_1K_1$	Z
0	00	0 1	0 1	0 1	0 d 1 d	0
0	01	1 1	1 1	1 0	1 d d 0	0
0	10	0 0	0 0	1 0	d 1 0 d	1
0	11	1 0	1 0	0 1	d 0 d 1	1
1	00	1 0	1 0	1 0	1 d 0 d	0
1	01	1 0	1 0	1 1	1 d d 1	0
1	10	1 1	1 1	0 1	d 0 1 d	1
1	11	0 1	0 1	1 0	d 1 d 0	0

x	y_2y_1	$y_2^{(n+1)} y_1^{(n+1)}$	D_2D_1	T_2T_1	$J_2K_2J_1K_1$	Z
0	0 0	0 1	0 1	0 1	0 d 1 d	0
0	0 1	1 1	1 1	1 0	1 d d 0	0
0	1 0	0 0	0 0	1 0	d 1 0 d	1
0	1 1	1 0	1 0	0 1	d 0 d 1	1
1	0 0	1 0	1 0	1 0	1 d 0 d	0
1	0 1	1 0	1 0	1 1	1 d d 1	0
1	1 0	1 1	1 1	0 1	d 0 1 d	1
1	1 1	0 1	0 1	1 0	d 1 d 0	0

(2) D Flip-flop

$$Z = \overline{x}y_2 + \overline{y_1}y_2 = \overline{xy_1y_2}$$

$$D_2 = \overline{x}y_1 + x\overline{y_1} + x\overline{y_2} = \overline{xy_1} + \overline{xy_1} + y_1\overline{y_2}$$

$$D_1 = \overline{x}y_2 + xy_2$$

x	$y_2 y_1$	$y_2^{(n+1)} y_1^{(n+1)}$	$D_2 D_1$	$T_2 T_1$	$J_2 K_2 J_1 K_1$	Z
0	0 0	0 1	0 1	0 1	0 d 1 d	0
0	0 1	1 1	1 1	1 0	1 d d 0	0
0	1 0	0 0	0 0	1 0	d 1 0 d	1
0	1 1	1 0	1 0	0 1	d 0 d 1	1
1	0 0	1 0	1 0	1 0	1 d 0 d	0
1	0 1	1 0	1 0	1 1	1 d d 1	0
1	1 0	1 1	1 1	0 1	d 0 1 d	1
1	1 1	0 1	0 1	1 0	d 1 d 0	0

T flip-flop

$$Z = xy_2 + \overline{y_1}y_2 = \overline{xy_1}y_2$$

$$T_2 = \overline{xy_2} + \overline{xy_1} + \overline{y_2}y_1 + \overline{xy_1}y_2$$

$$T_1 = \overline{\overline{xy_2}y_1} + \overline{xy_2}y_1 + \overline{xy_2}\overline{y_1} + \overline{xy_2}y_1 = x \oplus y_2 \oplus \overline{y_1}$$

x	$y_2 y_1$	$y_2^{(n+1)} y_1^{(n+1)}$	$D_2 D_1$	$T_2 T_1$	$J_2 K_2 J_1 K_1$	Z
0	0 0	0 1	0 1	0 1	0 d 1 d	0
0	0 1	1 1	1 1	1 0	1 d d 0	0
0	1 0	0 0	0 0	1 0	d 1 0 d	1
0	1 1	1 0	1 0	0 1	d 0 d 1	1
1	0 0	1 0	1 0	1 0	1 d 0 d	0
1	0 1	1 0	1 0	1 1	1 d d 1	0
1	1 0	1 1	1 1	0 1	d 0 1 d	1
1	1 1	0 1	0 1	1 0	d 1 d 0	0

JK flip-flop

$$Z = \overline{x}y_2 + \overline{y_1}y_2 = \overline{xy_1y_2}$$

$$J_2 = x + y_1$$

$$K_2 = x \oplus \overline{y_1}$$

$$J_1 = x \oplus \overline{y_2}$$

$$K_1 = x \oplus y_2$$

So....